

Lab 4: Normal-Normal Conjugacy, Posterior Predictive Distribution and Monte Carlo

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1. Normal-normal posterior for coffee-bag fill weights

A coffee-roasting company fills bags labeled as containing 500 grams. An inspector measures the fill weight of 12 bags:

```
y <- c(503, 498, 501, 505, 499, 502,  
       500, 504, 497, 501, 503, 499)  
  
n <- length(y)  
ybar <- mean(y)
```

Let θ be the unknown mean fill weight. Assume

$$Y_i \mid \theta \stackrel{\text{ind}}{\sim} \mathcal{N}(\theta, \sigma^2), \quad \sigma = 4,$$

and

$$\theta \sim \mathcal{N}(m_0, \tau_0^2), \quad m_0 = 500, \quad \tau_0 = 6.$$

The normal-normal posterior is

$$\theta \mid y \sim \mathcal{N}(m_n, v_n),$$

where

$$v_n = \left(\frac{n}{\sigma^2} + \frac{1}{\tau_0^2} \right)^{-1}, \quad m_n = v_n \left(\frac{n\bar{y}}{\sigma^2} + \frac{m_0}{\tau_0^2} \right).$$

Exercise 1: Posterior calculations

Question 1

Calculate n and $\bar{y}$.

```
cat("n =", n, "\n")
```

$n = 12$

```
cat("ybar =", ybar)
```

$ybar = 501$

Question 2

Calculate the data precision, prior precision, posterior precision, m_n , and v_n .

$$\begin{aligned}\text{data precision} &= \frac{n}{\sigma^2} = \frac{12}{16} = 0.75 \\ \text{prior precision} &= \frac{1}{\tau_0^2} = \frac{1}{36} = 0.0278 \\ \text{posterior precision} &= \frac{n}{\sigma^2} + \frac{1}{\tau_0^2} = 0.7778 \\ v_n &= \left(\frac{n}{\sigma^2} + \frac{1}{\tau_0^2} \right)^{-1} = (0.7778)^{-1} = 1.2857 \\ m_n &= v_n \left(\frac{n\bar{y}}{\sigma^2} + \frac{m_0}{\tau_0^2} \right) = 1.2857 \left(\frac{12(501)}{16} + \frac{500}{36} \right) = 500.964\end{aligned}$$

Question 3

Write the complete posterior distribution of $\theta \mid y$.

$$\theta \mid y \sim \mathcal{N}(m_n, v_n) = \mathcal{N}(500.964, 1.2857)$$

Question 4

Calculate the posterior mean, posterior variance, posterior standard deviation, and central 95% credible interval.

$$\text{posterior mean} = m_n = 500.964$$

$$\text{posterior variance} = v_n = 1.2857$$

$$\text{posterior SD} = \sqrt{v_n} = 1.1339$$

$$\begin{aligned} 95\% \text{ Credible Interval} &= m_n \pm 1.96\sqrt{v_n} \\ &= 500.964 \pm 1.96(1.1339) \\ &= 500.964 \pm 2.2224 \\ &= (498.742, 503.187) \end{aligned}$$

Question 5

Calculate $\Pr(\theta > 502 \mid y)$.

```
1 - pnorm(502, mean = 500.964, sd = sqrt(1.2857))
```

```
[1] 0.18
```

So $\Pr(\theta > 502 \mid y) = 0.18$.

Exercise 2: Density, CDF, and quantiles

Question 1

To calculate the posterior density evaluated at 502 grams, I use the mean and variance I found before and use the normal distribution function in R:

```
m_n <- 500.964
v_n <- 1.2857
sigma <- 4
dnorm(502, mean = m_n, sd = sqrt(v_n))
```

```
[1] 0.232
```

$$f_{\theta|y}(502) = 0.23$$

This density value is not the probability that $\theta = 502$, because density values are just not probabilities. It comes from integrating over an interval. The $Pr(\theta = 502) = 0$ because it is just a point on the pdf.

Question 2

Calculate the posterior CDF at 502 grams:

```
pnorm(502, mean = m_n, sd = sqrt(v_n))
```

```
[1] 0.82
```

$$Pr(\theta \leq 502 \mid y) = 0.82$$

Question 3

Calculate

$$Pr(499 < \theta < 502 \mid y)$$

using two CDF values.

$$Pr(499 < \theta < 502 \mid y) = F_{\theta|y}(502) - F_{\theta|y}(499)$$

```
pnorm(502, mean = m_n, sd = sqrt(v_n)) - pnorm(499, mean = m_n, sd = sqrt(v_n))
```

```
[1] 0.778
```

And so $Pr(499 < \theta < 502 \mid y) = 0.78$.

Question 4

Use `qnorm()` to calculate the 2.5th percentile, 50th percentile, and 97.5th percentile of the posterior.

```
q <- qnorm(c(0.025, 0.5, 0.975), mean = m_n, sd = sqrt(v_n))
cat("2.5th percentile =", q[1], "\n")
```

2.5th percentile = 499

```
cat("50 th percentile =", q[2], "\n")
```

50 th percentile = 501

```
cat("97.5th percentile =", q[3], "\n")
```

97.5th percentile = 503

Question 5

The 2.5th and 97.5th percentiles here agree with the endpoints of the central 95% credible interval from Exercise 1 Q4: (498.742, 503.187).

2. Monte Carlo estimation and Bayesian decisions

For posterior draws $\theta^{(1)}, \dots, \theta^{(M)}$,

$$\mathbb{E}[g(\theta) \mid y] \approx \frac{1}{M} \sum_{m=1}^M g(\theta^{(m)}).$$

Generate posterior draws:

```
set.seed(402)

M <- 10000

theta_draws <- rnorm(M, mean = m_n, sd = sqrt(v_n))
```

Exercise 3: Monte Carlo estimate of $g(\theta)$

Question 1

To define

$$g(\theta) = (\theta - 500)^2$$

It is the squared deviation of θ from 500.

And to calculate the Monte Carlo estimator of

$$\begin{aligned}\mathbb{E}[g(\theta) \mid y] \\&= \frac{1}{M} \sum_{m=1}^M g(\theta^{(m)}) \\&= \frac{1}{M} \sum_{m=1}^M (\theta^{(m)} - 500)^2\end{aligned}$$

```
mean((theta_draws - 500)^2)
```

```
[1] 2.2
```

= 2.20.

Question 2

Use the posterior draws to estimate $\mathbb{E}(\theta \mid y)$ and $\Pr(\theta > 502 \mid y)$.

```
cat("E(theta | y) =", mean(theta_draws), "\n")
```

```
E(theta | y) = 501
```

```
cat("Pr(theta > 502 | y) =", mean(theta_draws > 502), "\n")
```

```
Pr(theta > 502 | y) = 0.18
```

Exercise 4: Loss functions and Bayesian estimators

For this exercise, use

$$\phi \mid y \sim \text{Beta}(1, 3).$$

Under absolute error loss,

$$L_1(a, \phi) = |a - \phi|.$$

Under squared error loss,

$$L_2(a, \phi) = (a - \phi)^2.$$

The posterior expected loss of a candidate a is

$$R(a) = \mathbb{E}[L(a, \phi) \mid y].$$

In practice, posterior draws allow us to estimate this expected loss by

$$\hat{R}(a) = \frac{1}{M} \sum_{m=1}^M L(a, \phi^{(m)}).$$

The posterior is strongly right-skewed, so its mean and median are visibly different. This lets us see the difference between squared-error and absolute-error decisions. For a normal posterior, the posterior mean and median coincide.

Question 1

Generate 10,000 posterior draws and calculate the posterior mean and posterior median.

```
M <- 10000
post <- rbeta(M, shape1 = 1, shape2 = 3)
cat("posterior mean =", mean(post), "\n")
```

```
posterior mean = 0.252
```

```
cat("posterior median =", median(post), "\n")
```

posterior median = 0.207

Question 2

For candidate estimates $a \in [0, 1]$, calculate the empirical posterior expected absolute loss and squared loss:

$$\widehat{R}_1(a) = \frac{1}{M} \sum_{m=1}^M |a - \phi^{(m)}|,$$

$$\widehat{R}_2(a) = \frac{1}{M} \sum_{m=1}^M (a - \phi^{(m)})^2.$$

```
a_grid <- seq(0, 1, by = 0.001)
R1 <- sapply(a_grid, function(a) mean(abs(a - post)))
R2 <- sapply(a_grid, function(a) mean(abs(a - post)^2))
```

Question 3

First plot the pointwise absolute loss

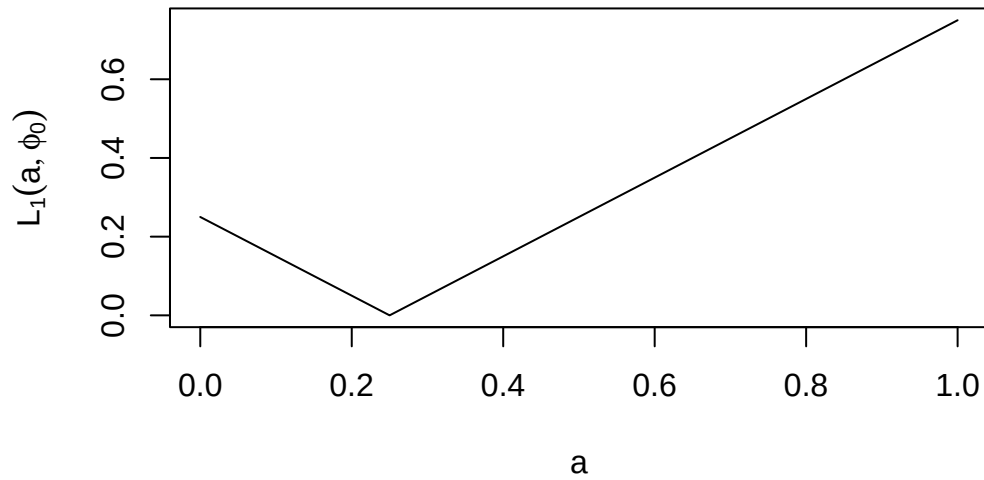
$$L_1(a, \phi_0) = |a - \phi_0|$$

for $\phi_0 = 0.25$. Then plot the posterior expected absolute-loss curve and add vertical lines at the posterior mean and posterior median.

```
phi0 <- 0.25
L1 <- abs(a_grid - phi0)

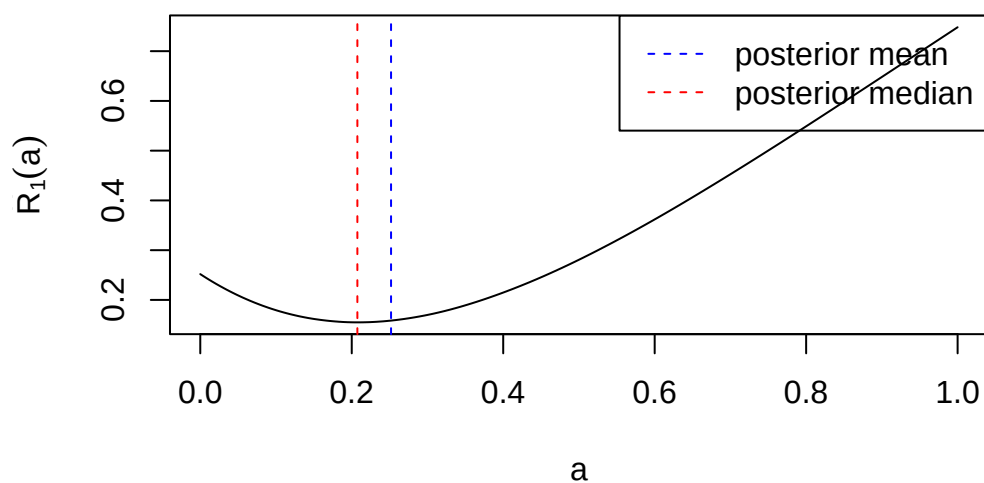
plot(a_grid, L1, type = "l",
     xlab = "a",
     ylab = expression(L[1](a, phi[0])),
     main = "Pointwise Absolute Loss, phi0 = 0.25")
```

Pointwise Absolute Loss, $\phi_0 = 0.25$



```
plot(a_grid, R1, type = "l",
     xlab = "a",
     ylab = expression(hat(R)[1](a)),
     main = "Posterior Expected Absolute Loss")
abline(v = mean(post), col = "blue", lty = 2)
abline(v = median(post), col = "red", lty = 2)
legend("topright", legend = c("posterior mean", "posterior median"),
     col = c("blue", "red"), lty = 2)
```

Posterior Expected Absolute Loss



Question 4

Find the value of a that minimizes the empirical absolute posterior loss and compare the absolute loss at the posterior mean and posterior median.

```
a_min <- a_grid[which.min(R1)]  
cat("a* (minimizer) =", a_min, "\n")
```

```
a* (minimizer) = 0.207
```

```
cat("R1(a*) =", min(R1), "\n")
```

```
R1(a*) = 0.155
```

```
cat("R1(post_mean) =", mean(abs(mean(post) - post)), "\n")
```

```
R1(post_mean) = 0.158
```

```
cat("R1(post_median) =", mean(abs(median(post) - post)), "\n")
```

```
R1(post_median) = 0.155
```

Here, we see that the value of a that minimizes the empirical absolute posterior loss is 0.207, and $R1(a^*)$ and $R1(\text{post_median})$ both give 0.155. The posterior median is the minimizer. The posterior mean gives a higher expected absolute loss.

Question 5

Find the value of a that minimizes the empirical squared posterior loss. Verify that it is close to the posterior mean.

```
a_min2 <- a_grid[which.min(R2)]  
cat("a* (minimizer) =", a_min2, "\n")
```

```
a* (minimizer) = 0.252
```

```
cat("R2(a*) =", min(R2), "\n")
```

```
R2(a*) = 0.0379
```

```
cat("R2(post_mean) =", mean((mean(post) - post)^2), "\n")
```

```
R2(post_mean) = 0.0379
```

Yep! It is close (basically the same) as the posterior mean.

3. Posterior predictive distribution

Let $\tilde{Y}$ be the fill weight of one new coffee bag. Conditional on θ ,

$$\tilde{Y} \mid \theta \sim \mathcal{N}(\theta, \sigma^2), \quad \theta \mid y \sim \mathcal{N}(m_n, v_n),$$

and

$$\tilde{Y} \mid y \sim \mathcal{N}(m_n, v_n + \sigma^2).$$

4. Monte Carlo from the posterior predictive distribution

The two-stage simulation is

$$\theta^{(m)} \sim p(\theta \mid y), \quad \tilde{Y}^{(m)} \mid \theta^{(m)} \sim \mathcal{N}(\theta^{(m)}, \sigma^2).$$

In R:

```
set.seed(402)

theta_draws <- rnorm(M, mean = m_n, sd = sqrt(v_n))

y_new_draws <- rnorm(M, mean = theta_draws, sd = sigma)
```

Exercise 5: Analytic posterior predictive calculations

Question 1

The complete posterior predictive distribution of $\tilde{Y} \mid y$:

$$\tilde{Y} \mid y \sim \mathcal{N}(500.964, 17.2857)$$

Question 2

The posterior predictive mean, variance, and standard deviation:

$$\begin{aligned}\mathbb{E}(\tilde{Y} \mid y) &= m_n = 500.964 \\ \text{Var}(\tilde{Y} \mid y) &= v_n + \sigma^2 = 1.2857 + 16 = 17.2857 \\ SD &= \sqrt{v_n + \sigma^2} = \sqrt{17.2857} \approx 4.1576\end{aligned}$$

Question 3

A central 95% posterior predictive interval:

$$\begin{aligned}\text{95\% Predictive Interval} &= m_n \pm 1.96\sqrt{v_n + \sigma^2} \\ &= 500.964 \pm 1.96(4.1576) \\ &= 500.964 \pm 8.1489 \\ &= (492.815, 509.113)\end{aligned}$$

Question 4

Calculate

$$\Pr(\tilde{Y} > 505 \mid y)$$

```
1 - pnorm(505, mean = m_n, sd = sqrt(v_n + sigma^2))
```

So $\Pr(\tilde{Y} > 505 \mid y) = 0.166$.

Question 5

Calculate the predictive quantiles at probabilities 0.025, 0.50, and 0.975.

```
q1 <- qnorm(c(0.025, 0.50, 0.975), mean = m_n, sd = sqrt(v_n + sigma^2))  
cat("2.5th percentile =", q1[1], "\n")
```

2.5th percentile = 493

```
cat("50 th percentile =", q1[2], "\n")
```

50 th percentile = 501

```
cat("97.5th percentile =", q1[3], "\n")
```

97.5th percentile = 509

Exercise 6: Monte Carlo estimator from the posterior predictive distribution

Question 1

Use the predictive draws to estimate the predictive mean, variance, standard deviation, probability above 505 grams, and central 95% predictive interval.

```
cat("predictive mean =", mean(y_new_draws), "\n")
```

predictive mean = 501

```
cat("predictive variance =", var(y_new_draws), "\n")
```

predictive variance = 17.4

```
cat("predictive SD =", sd(y_new_draws), "\n")
```

predictive SD = 4.17

```
cat("Pr(Ynew > 505) =", mean(y_new_draws > 505), "\n")
```

Pr(Ynew > 505) = 0.159

```
cat("95% predictive interval = (", quantile(y_new_draws, c(0.025, 0.975)), ")")
```

95% predictive interval = (493 509)

Question 2

To distinguish between estimating $\mathbb{E}[g(\theta) | y]$ using posterior draws and estimating $\mathbb{E}[h(\tilde{Y}) | y]$ using posterior predictive draws, we know $\theta^{(m)}$ draws only has uncertainty about the unknown mean θ after seeing the data; $\tilde{Y}^{(m)}$ draws carries two layers of uncertainty, including parameter uncertainty and the noise ϵ . So in code, `y_new_draws` uses `theta_draws` as an argument.

Exercise 7: A general function of a future observation

Define

$$h(\tilde{y}) = \mathbb{I}(\tilde{y} > 505).$$

Question 1

Define h in R and calculate the Monte Carlo estimator of

$$\mathbb{E}[h(\tilde{Y}) | y].$$

```
h <- function(y) y > 505  
mean(h(y_new_draws))
```

```
[1] 0.159
```

$\mathbb{E}[h(\tilde{Y}) \mid y] = 0.159.$

Question 2

Monte Carlo estimator is also an estimate of the posterior predictive probability $\Pr(\tilde{Y} > 505 \mid y)$, because for an indicator function, the expectation is a probability since:

$$\mathbb{E}[\mathbb{I}(\tilde{Y} > 505) \mid y] = 1 \cdot \Pr(\tilde{Y} > 505 \mid y) + 0 \cdot \Pr(\tilde{Y} \leq 505 \mid y) = \Pr(\tilde{Y} > 505 \mid y).$$